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ARTICLE

Enhancement of Lithium Transport by Controlling the Mesoporosity of Silica Monoliths filled by Ionic Liquids

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A systematic study of the effect of the mesopore diameter of silica monoliths with hierarchical porosity (meso-/macroporosity) on charge transport of confined ionic liquid with lithium salt is reported. Different measurements have been performed including ionic conductivity, Li^+ transference number, and ^7Li MAS NMR. For a concentration of 0.5 M of Li bis-trifluorosulfonylimide an optimum of ion transport is found for N-methyl,N-propylpyrrolidinium bis-trifluorosulfonylimide confined within silica mesopore of 10 nm diameter. For this value of mesopore diameter, the ionogel monolith shows better charge transport behaviour than the same, non confined ionic liquid. Confining ionic liquid within silica monoliths with hierarchical porosity (meso-/macroporosity) allowed also to evidence that macropores has no effect on ionic transport. The optimum of mesoporosity of silica for Li^+ diffusivity seems to be related to a good compromise between the amount of Li^+ interacting with the surface silanols and the amount of Li^+ in the volume of the pores. Smaller mesopores should give rise to too strong interactions of Li^+ with surface silanols wheras larger mesopores should lead to Li^+ strongly interacting with bis-trifluorosulfonylimide anions in the volume of the pores.

Introduction

The bottleneck for designing all solid and micro devices revolves around the use of solid electrolytes. Such devices are required to be leakage-free as well as to withstand the exposure to operating temperatures in a range around -20°C up to 100 °C. To date, the most commonly used solid electrolytes are based on LiPON, which displays low conductivity, and thus renders them poorly competitive. Liquid- or gel-type electrolytes with much higher conductivities than LiPON have been tested, none of which, however, are able to adequately fulfill the requirements of a leakage-free device, nor can they withstand the temperature required. Standard carbonate based electrolytes exhibit good ionic conductivities but they are liquid and volatile and are therefore impractical for the features required. Unlike solvent-based electrolytes, some ionic liquids (ILs) show wide electrochemical windows and fair ionic conductivities (1 to 10 $mS.cm^{-1}$ at 25 °C).[1] Moreover, ILs exhibit good chemical and thermal stabilities over a large temperature range and their vapor pressure is hardly measurable.[2] The bis-trifluoromethanesulfonylimide anion (TFSI) is often used in electrochemical devices, owing to its electrochemical stability toward; for high energy densities applications, the TFSI anion can be associated with pyrrolidinium cations, *e.g.* N-propyl,N-methylpyrrolidinium (Pyr13), resulting in a wide electrochemical stability. Nevertheless, for power applications, higher ionic conductivities will be required. Despite the noteworthy properties of ILs, leakage still remains an issue with respect to their use in all solid devices since they are in a liquid state at room temperature. It is for this reason that research efforts are being concentrated on obtaining a solid-like electrolyte with properties as close as possible to those of pristine ILs.

For such a purpose, the two principal approaches are: (i) to mix pre-formed inorganic particles with ILs to obtain a physical gel; (ii) to swell polymers or inorganic monoliths with ILs; and (iii) to polycondense or polymerize the confining network within these ILs directly, thus obtaining chemical gels. Whereas physical gels are crushable and present lower amounts of ILs, chemical gels, which are monolithic (*i.e.* ionogels), confine larger amounts of ILs in order to better preserve and even enhance the dynamics of the liquid state of the ILs.[3,4] Ionogels can be of different types depending on the nature of the matrix, which can be either polymeric, inorganic or hybrid organic-inorganic.[5,6] This ionogel approach allows the solid electrolyte to simultaneously act as both an ionic conductor and a separator in EDLCs or batteries.[7]

Nevertheless, for this last case, addition of lithium to the IL is known to increase its viscosity and decrease its conductivity. Studies on viscosity of confined ILs by means of physical apparatus external to the material have been reported.[8] The Walden product has been used for years to illustrate this conductivity-viscosity relationship.[9–13] However, the Walden product represents a macroscopic approach: a more

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microscopic approach based on the possible cation-anion association, from simple pairs up to nano-aggregated strongly interacting domains, has been proposed. This could be quantified by the ionicity of ILs, which is evaluated by the ratio between the molar conductivity measured by complex impedance spectroscopy (CIS) (macroscopic level) and by NMR (molecular level) ($\Lambda_{CIS}/\Lambda_{NMR}$).[14] NMR measurement probes the diffusion of each ion, associated or not with another, whereas CIS, which implies an electric field, probes the diffusion of each charged species, mostly single ions, and insignificantly associated ions. Thus the $\Lambda_{CIS}/\Lambda_{NMR}$ ratio may give information on the fraction of non-associated cations and anions.[14,15]

It remains a challenge to reduce ILs' viscosities and increase their conductivities without addition of any volatile compound. We have shown that confining ILs in silica based ionogels allows furthering this goal as well as offering a route to all-solid devices.[3] It has been shown that a higher conductivity of ionogels from 0.05 to 1.5 ms.cm^{-1}) was obtained with larger amount of IL used in the synthesis, which in the same time leads to an increase of the mesopore size inside the ionogel. The higher conductivity has been attributed to the higher volume of IL, but the effect of the size of the mesopores in ionogels has never been explored due to the difficulty to precisely control it.

In this study, we are exploring the influence of the mesopore size on ionic transport, by impregnating monolithic mesoporous silica with controlled mesopores diameter with IL. In order to measure conductivity the mesoporous silica should be in monolithic form. We have chosen to study, as supports for IL, monoliths with hierarchical porosity (macro-/mesoporous)[16,17] as they possess adjustable mesopore diameter and homogeneous flow through macropores, which insure efficient mass transport of molecules in catalysis, adsorption and chromatographic applications. The effect of the presence of macropores on ions transport will be examined by comparison with silica ionogels featuring only mesoporosity. The effect of mesopores size on the ionic transport will be conducted with silica monoliths featuring mesopore diameters between 3.7 and 20 nm. Besides overall conductivity measurements, the ionic transport properties of Li^+ in IL impregnated in silica monoliths featuring different mesopore sizes has been studied by lithium cation transference numbers measured by D.C. polarization and ^7Li MAS NMR. It is evidenced that an optimum pore diameter, for the IL studied, appears, showing the best ionic transport properties, tentatively interpreted in terms of fragility.

Experimental

Silica synthesis

Silica monoliths synthesis were adapted from NH_4OH[17] and MCM-41[16]. First, tetraethylorthosilicate (TEOS, Aldrich) (20 g) is left at -19°C for 1 h. In parallel, water (24.560 g) and then (2.313 g) nitric acid (68%) are added in a 100 mL Erlenmeyer. The mixture is stirred 5 min at room temperature. Polyethylene oxide (PEO 20 kDa) (2.534 g) is added to the mixture and stirred at room temperature until having

Table 1. D_{meso}, V_{meso}, V_{otal}, S_{BET}, V_{macro} and %wt of Pyr13-Li-TFSI adsorbed for different silica monoliths.

samples	D_{meso} / nm	V_{meso}/ mL g^{-1}	V_{total}/ mL g^{-1}*	S_{BET}/ m² g^{-1}	V_{macro}/ mL g^{-1} **	Pyr_{13}-Li-TFSI/ wt%
6599C-MCM41	3.7	0.89	3.3	1000	2.4	83
6640C-40°C-8h	6.5	1.06	3.5	756	2.4	83
6640C-40°C-12h	8	1.01	3.4	708	2.4	82
6654C-40°C-24h	10	1.20	3.7	700	2.5	85
6971C-50°C-24h	12	1.23	3.3	634	2.1	83
6971C-60°C-24h	14	1.25	3.6	545	2.3	84
RBX80C-80°C-24h	20	1.33	4.4	426	3.1	87

* determined from Electrolytic Solution (LiTFSI/Pyr13TFSI) adsorption
** $V_{macro} = V_{total} - V_{meso}$

complete dissolution of the polymer. The mixture is left 10 min at -19°C in the freezer to cool down the solution without freezing. The flask is then placed in an ice bath and stirred. TEOS (coming from the freezer) is directly added to the slurry and the solution is stirred for 30 min. The final composition of the mixture in molar ratio is: 1 Si / 0.60 EO unit / 0.26 HNO_3 / 14.21 H_2O. Polyvinyl chloride (PVC) tubes of 8 mm diameter and 10 cm length are closed on one side with a cap, sealed with parafilm and kept at -19°C in the freezer. The tubes are taken from the freezer and filled with the mixture of the ice bath. The tubes are then closed by caps and sealed with parafilm and left in water bath at 40°C for 3 days. The phase separation and the sol-gel process take place during this time. Monoliths are then removed from the tube molds and placed in 1 L water bath at room temperature. Water is changed every 30 minutes until reaching a neutral pH. The macroporous network is formed after this first step in acidic medium. Then the monoliths are submitted to basic treatments to create the mesoporosity with NH_4OH for mesopore diameters of 6 to 20 nm, or with NaOH in presence of cetyltrimethylammonium bromide (Aldrich) for mesopore size of 3.7 nm (MCM-41 type). For NH_4OH treatment, the monoliths from the first step are immersed in 1 L aqueous ammonia (0.1 M) in a polypropylene bottle and left in an oven at a temperature T (40 < T < 80°C) for different time t (8 < t < 24 h). The monoliths will be referred by their temperature and time treatment in Table 1. In a typical synthesis of MCM-41 like monolith, two wet monoliths of 10 cm (corresponding to 0.80 g each of dried and calcined SiO_2) are taken and placed at the bottom of an autoclave. An aqueous solution is prepared by dissolving 0.213 of NaOH in 192 g of water. Then, 5.831 g of CTAB are added and the mixture is stirred for 1 h at room temperature. The alkaline solution is poured on the autoclave containing the monoliths and let in an oven at 115 °C for 6 days. The final composition in molar ratio is: 1 Si / 0.60 CTAB /

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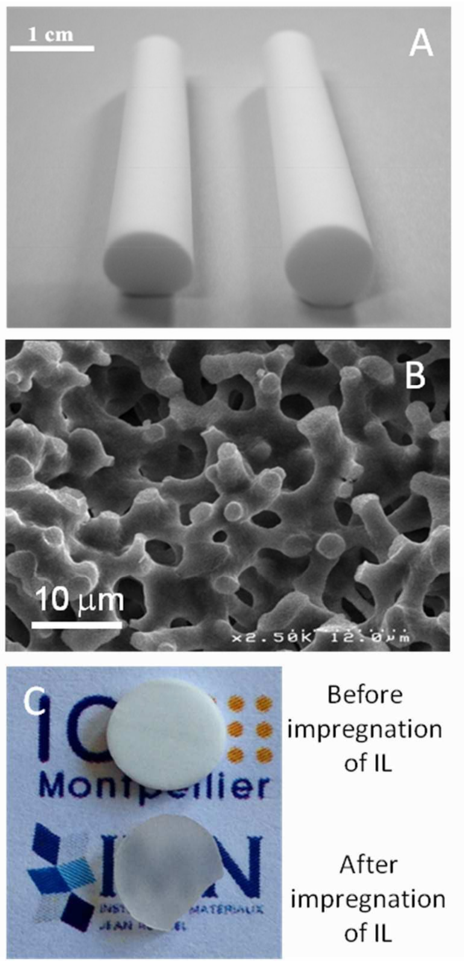

Fig 1. Photographies of silica monolith of 6 mm diameter and 7 cm length (A), SEM image of silica monolith with the macropores size of 4 microns (B) and slices (3 mm) of a silica monolith with the mesopores size of 10 nm before and after impregnation with ES (LiTFSI/Pyr13TFSI-IL)(C).

0.2 NaOH / 400 H_2O. The monoliths resulting of basic treatments are then placed in a water bath and water is changed every 30 min until neutral pH. The monoliths are then dried at room temperature during 4 days and calcined at 550°C for 8 h under air to totally remove remaining PEO. The resulting monoliths feature 6 mm diameter, 10 cm length and weight 80 mg/cm and a density of 0.28 g/cm^3.

Silica structural analysis

Monoliths were then cut precisely in 3 mm width pellets (6 mm diameter) with a saw for conductivity experiments. Silica monoliths present a hydroxylated surface and feature a SiOH density of ca. 3.7 SiOH/nm^2 evaluated by the weight loss of water by thermogravimetric analysis. The mesoporous textural properties of the materials were determined from the N_2 adsorption/desorption isotherms at 77 K measured on a Micromeritics Tristar 3000 apparatus. The samples were previously outgassed in vacuum at 250 °C for 12 h. The mesopore diameter was estimated using the desorption branch of the isotherm and using the Broekhoff and de Boer method, as it has been previously demonstrated as one of the most accurate method for mesopore size determination for hydroxylated silica materials.[18] The mesoporous volume was taken on the plateau just after the step in nitrogen adsorption, at the end of the capillary condensation. The macroporous network of the monoliths was studied using a Hitachi S-4500 I scanning electron microscope (SEM) at “IEM Montpellier Plateau Technique du Pole Chimie Balard”. Monoliths are formed by skeletons of 3 microns width and a homogeneous network of macropores of 4 microns, confirmed by mercury porosimetry. Macroporous volumes of some monoliths have been determined by mercury porosimetry on the courtesy of Dr. Benedicte Lebeau from Mulhouse, France. The total pore volumes of the monoliths were also assessed by weighing the amount of ionic liquid (electrolytic solution of ionic liquid with a volumic mass 1.4542 g.cm-^{3}, measured at 35 °C by means of Anton Paar DMA4100M density meter) adsorbed in the monoliths and substracting the mesoporous volume obtained by nitrogen adsorption. Similar macroporous volumes were obtained by the two technics (around 2 mL.g^{-1}) and the adsorption of ionic liquid was preferred for convenience.

Ionogels preparation

Electrolytic solution (ES) was obtained by dissolving bis(trifluorosulfonyl)imide lithium salt (LiTFSI, 0.1 g, 3M, 99%) in N-methyl-N-propylpyrrolidinium bis(trifluorosulfonyl)imide (Pyr13TFSI, 1 g, Solvionic, 99.5%). Monoliths were impregnated by immersion in an excess of ES during 24h at room temperature. Before measurements, samples were dried at 50°C under vacuum for 24h.

Electrochemical measurements

The ionic conductivities were determined by complex impedance spectroscopy (CIS) using a BioLogic VMP2-Multichanel Potentiostat by varying the temperature from -20 °C to 90 °C. The frequency range used for impedance measurements was 184 kHz - 20 mHz and the amplitude used was 7 mV. Lithium cation transference numbers (t_{Li+}) were determined with symmetric cell (metal Li/ionogel or ES/Li metal) and were measured using the D.C. polarization method. The frequency range for the CIS measurements was 184 kHz-100 mHz, with a 10 mV dc signal. Polarization was performed with a 20 mV dc signal. The lithium cation transference numbers were calculated as:

$$t_{Li^+} = \frac{I_s(\Delta V - R_0 I_0)}{I_0(\Delta V - R_s I_s)} \qquad (1)$$

Where ΔV is the polarization potential (20 mV); R_0 is the interfacial resistance of the lithium electrode before polarization; R_s is the interfacial resistance of the lithium electrode after polarization; I_0 is current at the start of polarization, and I_s is the steady-state current at polarization. ^7Li solid state static NMR experiments were carried out at room temperature on a Bruker Avance-200 spectrometer (B_0 = 4.7T, Larmor frequencies $\nu_0(^7Li)$ = 77.78 MHz). Pieces of ionic liquid confined in silica matrix were cut and placed into a cylindrical 2.5 mm o.d. zirconia rotor. ^7Li static NMR spectra were acquired by making use of a non selective single pulse sequence with $\pi/2$=2.3 µs coupled with a pre-acquisition time

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of 5 μs and a recycle time of 1 s. All spectra displayed in this work were normalized taking into account the number of scans, the received gain, and the mass of sample. ^{7}Li integrated intensities were determined by using spectral simulation.[19]

Results and discussion

Materials processing

In order to explore systematically the influence of the mesopore size on ionic transport, silica with well controlled mesopores diameter were impregnated with IL. Silica monoliths (6 mm diameter, 10 cm length) featuring hierarchical porosity (mesoporous/macroporous) have been synthesized by a combination of spinodal decomposition (between SiO_2/PEO and water) and sol-gel process in acidic medium to control the macroporosity, followed by basic treatment to create the mesoporosity (Fig. 1).[17] All monoliths feature macropores size of 4 microns and skeletons of 3 microns (Fig. 1). Macroporous volumes of all monoliths are around 2 mL.g^{-1} (Fig. 2, Table 1). Mesopores diameters of the monoliths were adjusted between 6 and 20 nm by changing the temperature and the duration of the basic treatment (Table 1). Additionally, a pseudomorphic transformation of the silica monolith into MCM-41 monolith was performed to reach smaller mesopore size of 3.7 nm from a synthesis adapted from ref. 16. Monoliths will be further named by their mesopores diameters. Slices of 3 mm of monoliths were cut precisely with a saw to obtained flat surface of pellets necessary to conductivity measurements. The monolithic pellets were then totally filled with the electrolytic solutions (ES). Figure 1C shows a 10 nm pores diameter silica monolith before and after impregnation. The silica monolith becomes transparent after impregnation, in accordance with the full filling of the mesoporosity and macroporosity by ES. The weight amount of ES within silica monolith after impregnation is presented in table 1. By using the density of ES, one can estimate the total pore volume of the monoliths and by substraction of the mesoporous volume determined by nitrogen sorption determine the macroporous volume. Good accordance was found between macropores volume determination by ES adsorption in comparison to mercury porosimetry (Fig. 2, Table 1). It is worth to point out that this result makes of ES or ionic liquid adsorption a fast technic for total volume (and therefore macroporous volume) determination of monoliths with no need of sample preparation and heavy technic as mercury porosimetry.

Study of the confined ionic liquid

The ionic conductivity in ES impregnated silica monoliths with different mesopores diameters has been conducted by complex impedance spectroscopy (CIS) (Fig. 3). CIS measurements gives overall ionic transport, *i.e.* a mean value of "bulk" (IL confined closer to the middle of the pores than to the interface) and interface phenomena. Fig. 3A shows the ionic conductivity plots for the different silica monoliths and for neat ES. The monoliths' conductivity calculations are made on the basis of the same form factor for each sample, *i.e.* they

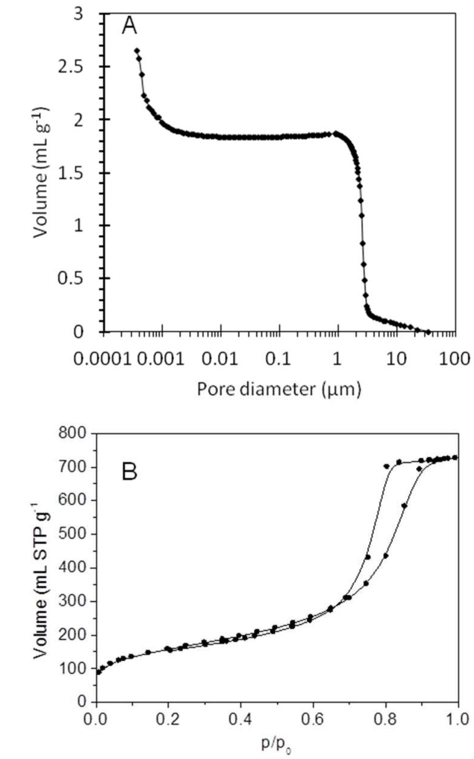

Fig 2. Typical Hg intrusion porosimetry (A) and nitrogen sorption isotherm at 77K (B) of silica monoliths (example of a silica monolith with 10 nm mesopore diameter).

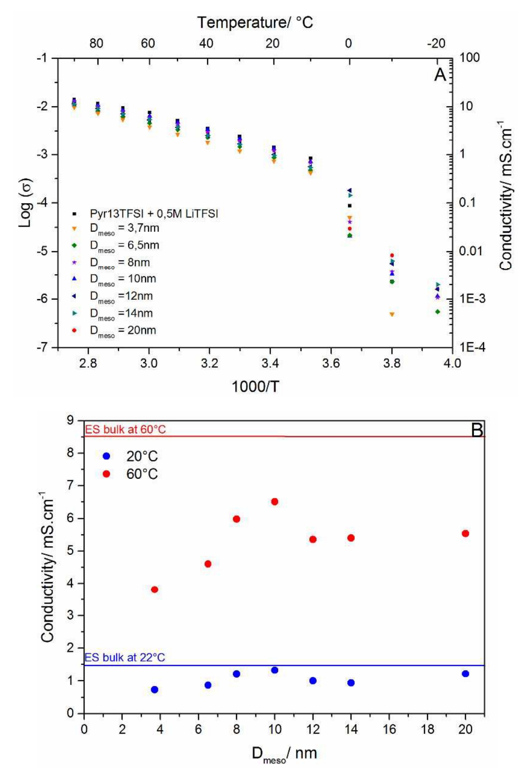

Fig 3. Conductivity plots Arrhenius for neat ES and ES impregnated monoliths of different mesopore diameters (A) and conductivity as a function of mesopore size at 22°C and 60°C (B).

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Table 2. σ_0, T_0, T_0/Tg and D for ES and the different monolith.

samples	Ln σ_0	T_0/ K	T_0/T_g*	D
Pyr_{13}-Li-TFSI	-7.6	185 +/- 10	0.97	3.4 +/- 0.7
D_{meso} = 3.7	-4.5	102 +/- 9	0.53	17.9 +/- 3
D_{meso} = 6.5	-7.1	168 +/- 4	0.88	5.1 +/- 0.4
D_{meso} = 8	-7	168 +/- 3	0.88	4.9 +/- 0.3
D_{meso} = 10	-8.1	196 +/- 1	1.01	2.7 +/- 0.1
D_{meso} = 12	-7	169 +/- 4	0.88	5.0 +/- 0.4
D_{meso} = 14	-7.2	178 +/- 7	0.94	4.2 +/- 0.6
D_{meso} = 20	-7.2	164 +/- 5	0.83	5.1 +/- 0.5

*Tg obtained by DSC (S.I. 2)

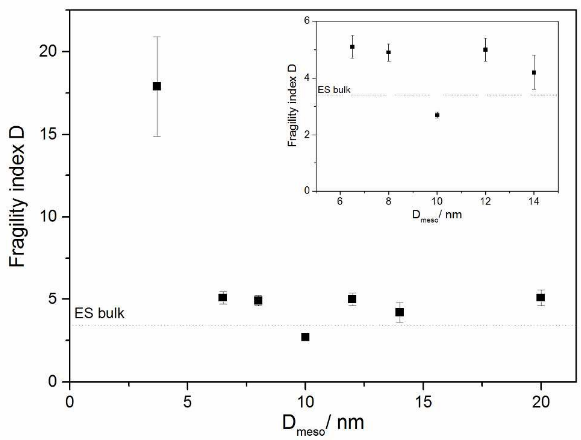

Fig.4 Fragility index (D) as a function of pore size.

correspond to lower amounts of ES in monoliths than for neat ES: consequently for monoliths, they do not correspond to the intrinsic conductivity of the confined ES but rather to the conductivity averaged in the whole monoliths. Nevertheless, between 10 and 90 °C, each impregnated monoliths shows a remarkable conductivity close to that of bulk ES. The striking result is the existence of a maximum of conductivity for an ES impregnated silica monolith with mesopore diameter of 10 nm (examples at 20 and 60 °C in Fig. 3B). This optimum has never been observed in other systems or silica ionogels. We show here for the first time the dependency of ionic transport with the mesopore size. As comparison, previously reported silica ionogels exhibit similar conductivity between 0.05 and 1.7 mS cm^{-1} at 20°C with similar mesopore size.[3] In this last case, the exact mesopore diameter of silica ionogels was difficult to evaluate as it has been determined after removing IL by solvent extraction, which can conduct to pore rearrangements.

However, this result highlights the fact that the macroporosity present in the monolith has no benefic effect on ions transport in contrary to what was observed for molecules in catalysis.[17] Only the mesopores of these materials influences the ionic transport.

The ionic conductivity follows the typical non-Arrhenius behavior and is well described by the Vogel-Tamman-Fulcher (VTF) equation (equation 2).

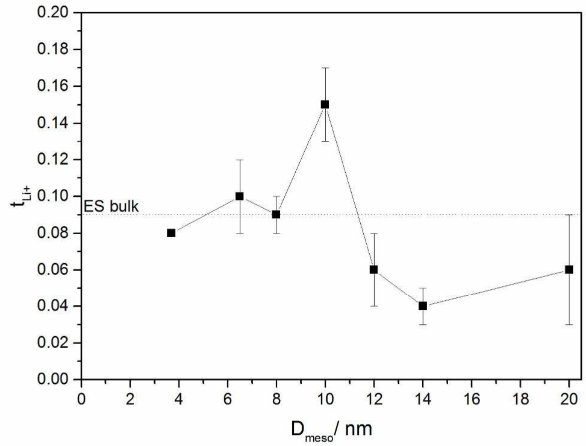

Fig 5. Lithium transference number (t_{Li+}) as a function of pore size.

$$\sigma = \sigma_0 \exp\left[-\frac{DT_0}{T-T_0}\right] \quad (2)$$

In this equation, σ_0 is the theoretical conductivity at infinite temperature, T_0 the ideal glass transition temperature, and D the fragility index; D is inversely proportional to the fragility of the liquid. The fragility index is related to the temperature dependence of the dynamical features of the liquid, influencing the transport properties such as conductivity and viscosity. A low D factor corresponds to a high fragility and a high ionic diffusion into the pores. Based on this model, σ_0, T_0, and D parameters were obtained (S.I. 1) and their values are summarized in Table 2. The fit was performed within a broad temperature range, in the liquid region, well above liquid-to-solid transition.

The fragility index D shows a minimum for the monolith with the mesopore diameter of 10 nm (Fig. 4), so the highest ionic diffusion is obtained for this mesopore diameter, accordingly to the conductivity measurements results (Fig. 3). Moreover, T_0/T_g is equal to 1 for the monolith featuring mesopores diameter of 10nm, accordingly with a fragile liquid[20] (Table 2; experimental glass transition temperature T_g was obtained from DSC measurements, see S.I. 2).

Lithium cation transference numbers (t_{Li+}) were measured using the D.C. polarization method with silica monoliths featuring different mesopores diameters. For large mesopore diameter (between 12 and 20 nm), t_{Li+} shows its lowest values, lower than that of bulk ES. For the smallest mesopore diameter (between 3.7 and 8 nm), t_{Li+} values are similar to that of bulk ES. A maximum value for t_{Li+} is obtained for the monolith with mesopore diameter of 10 nm, as for the other measurements.

^{7}Li NMR allowed investigating lithium environment within the ionogels (Fig. 6). The spectrums for bulk liquid ES as well as those for monoliths were recorded without MAS conditions. A sharp resonance, characteristic of liquid is observed at -0.5 ppm for bulk ES. Confinement within monoliths with the smallest pore sizes (high silica interface surface / lithium amount ratio) corresponds to a drastic shift toward higher

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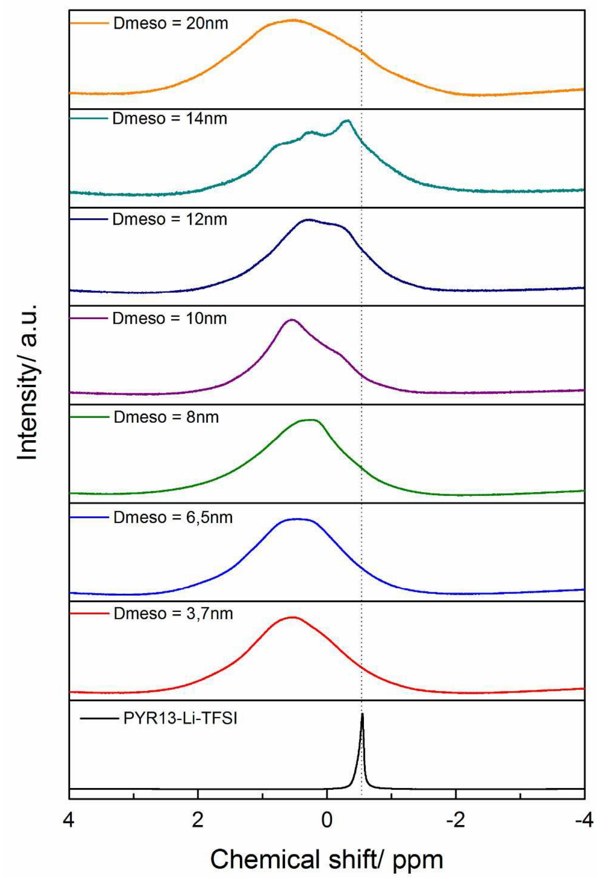

Fig.6 RT ^{7}Li static NMR spectra, for bulk ES and for silica monoliths with different mesopore diameters impregnated by ES.

frequencies : this was previously shown to be simultaneous with the de-coordination of TFSI anion from lithium cation,[3] to the benefit of close interaction between lithium cation and silica surface. This phenomenon is accompanied by a strong broadening of the NMR line, assigned both to the confinement of the liquid within the pores and to the overlapping of resonances due to a distribution of local lithium environments. Upon increasing the pore size, for a 10 nm size, NMR line features become more apparent and the contribution rising at lower frequencies, *i. e.* closer to that of bulk ES, is clearly visible.

From 10 nm to 14 nm pores sizes, the NMR spectra features are more visible, suggesting narrower individual contributions that may be the sign of an increased mobility of lithium ions. To go further would require an accurate deconvolution of the NMR spectra to determine the respective contribution of broad and narrow lines to the observed resonance. This is not possible here, due to the strong overlapping of the different contributions. Nevertheless contribution rising at lower frequencies for 10 nm pores seems to correspond to an increasing, when pore size increases above 10 nm, contribution of lithium ions interacting less with the silica interface, which could be due to saturation at 10 nm of the

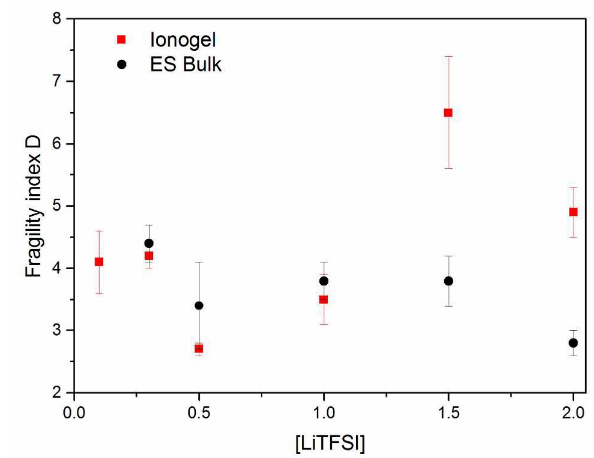

Fig. 7 Fragility index D as a function of [LiTFSI] for the monolith of 10 nm mesopore diameter impregnated with ES and comparison with bulk electrolytic solution.

silica interface by lithium ions. All NMR spectra show significant contributions at chemical shifts ranging from that of bulk ES up to that corresponding to the highest silica interface surface / lithium (or ES) amount ratio, obtained for the smallest pores size. It seems reasonable to consider the overall measured Li ions mobility properties as the combination of mobility properties corresponding to local Li environments, which gives rise to broad and narrow contributions, respectively.

Discussion

The preferential segregation of lithium cations at the silica interface is clearly evidenced by ^{7}Li MAS NMR measurements, as previously stated for ionogels.[3] We can here hypothesize that the density of surface silanols may also be a factor influencing cations diffusion. Nevertheless, in the present study for a silanol density of ~3.7 SiOH/nm^2, the main parameter is the mesopore size of the support. The dynamics of the silanol groups could help creating for the Li^+ cation a diffusion path at the interface, and the optimum pore diameter found for 10 nm might indicate that the best diffusion of ions should occur when (*i*) the silica interface is saturated by lithium ions and when (*ii*) an almost equal ratio of ions are present at the surface of the pore and in the volume of the pore. This assumption was also proposed in catalysis, where the best compromise between the highest reactivity and the highest diffusion was estimated for an equal amount of molecules in the surface of the pore and in the volume of the pore (leading to an optimal mesopore diameter equal to 5-7 times the size of the molecule, calculated from the Ruckenstein law).[21] It seems from this preliminary approach that the feature found for molecules could also be applicable for ions. Moreover, the optimal pore size determined experimentally here is fully consistent, although no exactly equal since confined ILs are different, with the dimensions found for instance with simulation,[22] with resonance shear measurement[23] and with combined broadband dielectric spectroscopy and pulsed field gradient NMR.[24]

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The index of fragility D has been explored as a function of LiTFSI salt concentration for the monolith with 10 nm mesopore diameter. The dependence of D value to the concentration of lithium salt, specifically when confined in a given pore size of 10 nm, presents its lowest value for a 0.5 M concentration, whereas D is rather constant in the bulk electrolytic solution (Fig. 7). It thus seems that the optimum conductivity / pore size that we obtain here corresponds to an optimum balance between surface of silica interface and the amount of confined IL with a specific lithium salt content.

Conclusions

The determining effect of the silica mesopore diameter to confined ionic liquid with lithium salt is evidenced here. The study of ionic transport of Li^+ in N-methyl,N-propylpyrrolidinium bis-trifluorosulfonylimide confined in silica monoliths with hierarchical porosity (meso-/macroporosity) with different mesoporosities has clearly evidenced different important points on ion transport: (i) macroporosity has no effect on ionic transport, (ii) an optimum of mesopore diameter exists, (iii) ion transport depends on Li^+ concentration and saturation of the interface, (iv) ionic transport would certainly depend on silanols density at the surface of the materials, (v) an equal number of ions at the surface of the pore and in the volume is suspected to be at the origin of an optimum ionic transport. In the case of the studied silica monoliths (with 3.7 $SiOH.nm^{-2}$) impregnated with N-methyl,N-propylpyrrolidinium bis-trifluorosulfonylimide, the optimum mesopores diameter was found at 10 nm for an optimum concentration of 0.5 M Li bis-trifluorosulfonylimide. For this value of mesopore diameter, the ionic liquid confined in silica monolith shows better lithium transport behaviour than bulk ionic liquid. Smaller mesopore size probably leads to stronger interaction of Li with the silica surface and decreases the transport. Higher mesopore size probably lead to a higher amount of lithium cation coordinated with bis-trifluorosulfonylimide anion, decreasing the overall transport of Li^+. Such a good balance between pore size and charge transport would have to be tune for any other, more or less bulky, ionic liquid, in order to optimize the features of the material. This striking result opens perspectives for systematic studies regarding effect of pore size for other values of lithium salt concentration with a given ionic liquid, as well as for a given salt concentration and other ionic liquids. As for other porous solids confining liquids, a best matching between porous host network and confined liquid should give the best macroscopic performances.

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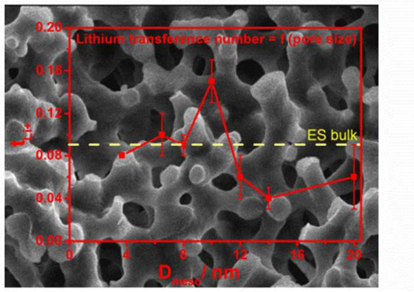
TOC entry



A systematic study of the effect of the mesopore diameter of silica monoliths with hierarchical porosity (meso-/macroporosity) on charge transport of confined ionic liquid with lithium salt shows an optimum for mesopore of 10 nm diameter.